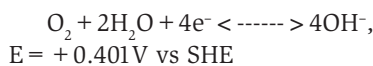


The oxidation of the iron electrode during the discharging cycle involves the following steps:

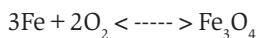
1. $\text{Fe} + 2\text{OH}^- \rightleftharpoons \text{Fe}(\text{OH})_2 + 2\text{e}^-$, E^0 (standard redox potential) = -0.88V vs. SHE (standard hydrogen electrode)
2. $\text{Fe} + 3\text{OH}^- \rightleftharpoons \text{HFeO}_2^- + \text{H}_2\text{O} + 2\text{e}^-$
3. $\text{HFeO}_2^- + \text{H}_2\text{O} \rightleftharpoons \text{Fe}(\text{OH})_2 + 2\text{OH}^-$
4. $\text{Fe}(\text{OH})_2 + \text{OH}^- \rightleftharpoons \text{FeOOH} + \text{H}_2\text{O} + \text{e}^-$, $E^0 = -0.56\text{V}$ vs SHE
5. $3\text{Fe}(\text{OH})_2 + 2\text{OH}^- \rightleftharpoons \text{Fe}_3\text{O}_4 + 4\text{H}_2\text{O} + 2\text{e}^-$, $E^0 = -0.66\text{V}$ vs SHE
6. $2\text{Fe}_3\text{O}_4 + 2\text{OH}^- \rightleftharpoons 3\text{Fe}_2\text{O}_3 + \text{H}_2\text{O} + 2\text{e}^-$, $E^0 = -0.60\text{V}$ vs SHE
7. $2\text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{H}_2 + 2\text{OH}^-$, $E^0 = -0.83\text{V}$ vs SHE

The reverse reactions represent the charging half cycle.

The discharge reaction at the positive air electrode involves four electrons as shown below:



The overall cell reaction is then:



Reactions 1 to 3 shown above are desired capacity-generating reactions in the iron electrode with a theoretical capacity of 960mAh g^{-1} . First, the iron is oxidised from metallic iron in the alkaline electrolyte to $2+$, then further oxidised to a $3+$ state. Further discharge converts iron to oxyhydroxide shown in reaction 4 and magnetite as per reaction 5, requiring higher overpotential.

Magnetite is hard to reduce to metallic iron and is less conductive than Fe metal. This increases the internal resistance of the cell, resulting in a large overpotential during charging.

Apart from these reactions, the evolution of hydrogen may also take place at open circuit voltage (OCV) because iron is unstable thermodynamically in an aqueous system. The standard hydrogen evolution reaction (HER) potential is 40mV posi-

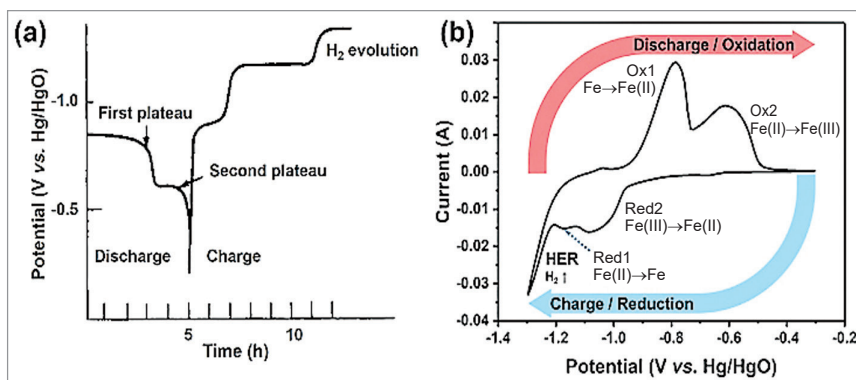


Fig. 6: (a) Charging/discharging curve, (b) CV curve of Fe-air battery

tive to the iron electrode's OCV. The iron electrode undergoes self-discharge due to the corrosion process and lowers the faradic efficiency and water loss in the system. Hence, overcharging and electrolyte top-up are necessary.

These problems can be overcome by adding small quantities of additives in the electrolyte that suppress hydrogen evolution and raise the overpotential of the iron electrode, decreasing the electrode resistance.

Fig. 6 shows a typical discharge/charge and CV curve of the iron anode in alkaline electrolyte.

The first and second plateaus shown in Fig. 6(a) are during the discharge cycle of a typical iron metal battery, corresponding to the oxidation of Fe^0 to Fe^{2+} and Fe^{2+} to Fe^{3+} , respectively. The cyclic voltammetry (CV) curve shows two pairs of major peaks at $-0.79/-1.17\text{V}$ (Ox1/Red1) and at $-0.62/-1.09\text{V}$ (Ox2/Red 2).

The first oxidation peak, i.e., Ox1, occurs during the formation of $\text{Fe}(\text{OH})_2$ as per reaction 1 shown earlier. The corresponding reduction peak, i.e., Red1, is accompanied by

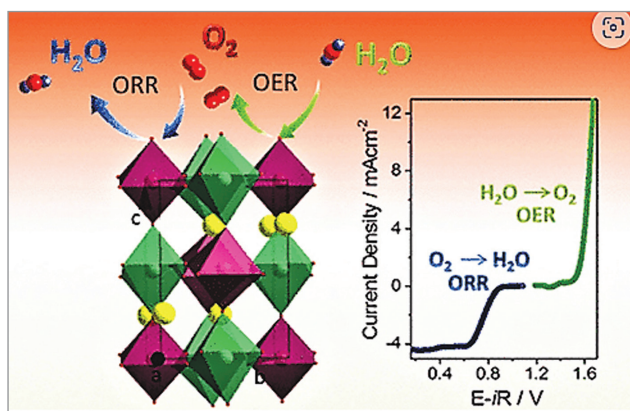


Fig. 7: Representation of OER and ORR in perovskite oxide

a steep peak indicating that parasitic hydrogen evolution reaction (HER) occurs. The second peaks (Ox2/Red2) correspond to complex transitions between Fe^{2+} and Fe^{3+} , forming FeOOH , Fe_3O_4 , Fe_2O_3 , etc. The generation of irreversible Fe_2O_3 causes passivation leading to capacity reduction and ultimately failure.

The reaction that generates hydrogen is not desirable as it limits the oxidation reaction, but it stops when Fe_3O_4 is formed. In fact, the electrode reactions at the iron electrode are complex and still a matter of speculation owing to solid-state transformation within the surface film, and it is thought that several intermediates occur during charging and discharging.

In simple terms, during the discharge of the battery, metallic iron is converted to magnetite (Fe_3O_4), and